



WORKSHOP ON LIQUID DEMOCRACY

URL: <https://democracynet.eu/activities/ldws2022/>

Date: December 15-16, 2022

Venue: University of Zurich

1. GENERAL INFORMATION

Grant number: IZSEZ0_214528

Starting date: 01.11.2022

Project title: Perspectives on liquid democracy

End date: 31.01.2023

Funded participants: Manon Revel (for details, please see appendix on p.19).

2. DESCRIPTION OF THE SCIENTIFIC EXCHANGE

2.1 WORKSHOP ACTIVITIES

The interdisciplinary workshop focused on liquid democracy, a democratic innovation that has gained increasing visibility over the past five years. It brought together a very diverse and interdisciplinary community of scholars and researchers to enable exchange among them, to identify the state of the art in the field, and to start a productive dialogue that can foster our understanding of liquid democracy, its concrete potential, and limitations, from complementary perspectives. The workshop addressed questions as diverse as:

- What is liquid democracy, what kind of concept is it and how can we best define it? Is there only one definition of liquid democracy, or can there be different conceptualizations?
- What is the logical domain of application of liquid democracy (private vs. political, party vs. government)?
- Which liquid democracy models are best suited to derive useful and realistic insights? Which are the mathematical challenges they pose?
- How can we best design software that helps us implement this idea? What are the technical challenges (from a computer science point of view) posed by this idea?
- How can liquid democracy be implemented within existing systems? What are the social, political, legal consequences of introducing this idea? How does liquid democracy impact participation?
- Why should we implement liquid democracy, if at all? Is it a legitimate form of decision-making? Which normative problems does it help solving?

2.2 RESULTS OF THE WORKSHOP

The goals of the international workshop on liquid democracy were a.) bringing together the interdisciplinary community of scholars and researchers, b.) identifying problems and limitations of liquid democracy, and c.) promoting and communicating the concept of liquid democracy to a broader audience. To that end, we used the funds of the scientific exchange grant to organize and host an interdisciplinary workshop on liquid democracy. Below, we present the final report on the liquid democracy workshop as our results for the scientific exchange grant and the overall output data for this scientific exchange grant. In addition to the written report, we would like to highlight the public roundtable session¹ that we recorded and published online.²

2.2.1 WORKSHOP PARTICIPANTS

Alessandro Checco (University of Rome La Sapienza)	Giovanni Davide Zenga (Polytechnic University of Milan)
Alois Paulin (University of Applied Sciences Ludwigsburg)	Gregory Butterworth (Cardiff University)
Andrei Constantinescu (ETH Zurich)	Henrieta Hajnali (University of Vienna)
Anson Kahng (University of Rochester)	Hugo Gilbert (Paris Dauphine University)
Bryan Alexander Ford (EPFL)	Jobst Heitzig (Potsdam Institute for Climate Impact Research)
Carlo Brancati (Digital entrepreneur and independent consultant)	Julia Wagner (ETH Zurich)
Catarina Pereira (University of Zurich)	Karsten Donnay (University of Zurich)
Chiara Valsangiacomo (University of Zurich)	Lucien Heitz (University of Zurich)
Christian Damsgaard Kristensen (Legal Consultant of the Danish Ministry of Defence)	Manon Revel (Massachusetts Institute of Technology)
Christina Isabel Zuber (University of Konstanz)	Miklovana Tuci (University of Zurich)
Cristina Sarasua (University of Zurich)	Nimrod Talmon (Ben-Gurion University)
Davide Grossi (University of Groningen)	Patryk Morawski (ETH Zurich)
Dimitrios Karoukis (CEO of Deliberative Technologies)	Rachael Colley (University of Toulouse)
Dylan Johnson (University of Hamburg)	Robin Fritsch (ETH Zurich)
Ehud Shapiro (Weizmann Institute of Science)	Sukmo Kim (University College London)
Marco Deseriis (Scuola Normale Superiore)	Theo Delemazure (Paris Dauphine University)
	Victoria Herbig (ETH Zurich)
	Yann Vonlanthen (ETH Zurich)
	Yepeng Ding (ETH Zurich)

¹ Information on the public roundtable: <https://democracynet.eu/video-liquid-democracy-roundtable/>

² Recording of the roundtable: <https://drive.google.com/file/d/1lDIW9AzaK0qoj0x-PxzPGIKjb9lPsx9H/view>

2.2.2 SCIENTIFIC EXCHANGE OUTPUT – PROGRAM OVERVIEW

<u>DAY 1: THURSDAY, DECEMBER 15, 2022</u> Institute of Political Science, Affolternstrasse 56, 8050, Zurich Room AFL-F-121 (Welcome session, panel and roundtable) Room AFL-F-172/73 (Poster session)	
11.00 – 11.30	Arrival of experts and participants
11.30-12.00	Welcome and introduction to the workshop
Poster presentations	
12.00-13.30	Hugo Gilbert, Rachael Colley and Theo Delemazure “Liquid Democracy - how critical are you in your network?” Jobst Heitzig “Non-majoritarian consensus-oriented liquid democracy? Combining probabilistic voting rules and flexible delegation systems in the vodle.it app” Sukmo Kim “Constitutive Referendums as Exceptional Democratic Act? Constituent Power, Referendums and the Case of Fusion” Christian D. Kristensen “Democracy, Decentralization, and Political Evolution: How Technology Enables the Liberal Democracy”
13.30-14.00	Coffee break (Cityport Mensa)
14.00-14.10	Presentation of DemocracyNet & sponsors
Panel I: Implementations of Liquid Democracy Chair: Cristina Sarasua	
14.10-14.50	Ehud Shapiro “Grassroots vs. Constitutional Liquid Democracy”
14.50-15.00	Break
15.00-15.40	Alois Paulin “Self-governance? Smart Anarchy? Beyond Bureaucracy? How Liquid Democracy could radically transform public governance”
15.40-16.20	Dimitrios Karoukis “Deliberative democracy with dilutive voting power sharing”
16.20-16.30	Wrap up and end of workshop’s day 1
16.30-18.00	Break (hotel check-in for participants)
Public roundtable “Liquid Democracy”	

18.00-18.45	<i>Moderated Roundtable with Karsten Donnay, Bryan Alexander Ford, Davide Grossi and Christina Isabel Zuber</i>
18.45-19.30	<i>Q&A with the public</i>
19.30 - 22.00	<i>Aperò riche with all participants and speakers</i>
22.00	<i>End of day 1</i>

<u>DAY 2: FRIDAY, DECEMBER 16, 2022</u> Department of Informatics, Binzmuehlestrasse 14, 8050 Zurich Room BIN-0-K12/13	
9.00-9.10	<i>Welcome and introduction to day 2</i>
Panel II: Political Philosophy and Economy of Liquid Democracy Chair: Manon Revel	
9.10-9.50	<i>Marco Deseriis "Is Liquid Democracy Compatible with Representative Democracy? Insights from the Experience of the Pirate Party Germany" *</i>
9.50-10.30	<i>Alessandro Checco "Online communities, misinformation, and post-truth - a computational social science perspective"</i>
10.30-10.45	<i>Break</i>
10.45-11.25	<i>Julia Wagner "Credence Policies and Experts in Democracy" *</i>
11.25-12.05	<i>Dylan Johnson "Electoral Market Structure under Liquid Democracy"</i>
12.05-14.00	<i>Lunch break</i>
Panel III: Mathematics of Liquid Democracy Chair: Chiara Valsangiacomo	
14.00-14.40	<i>Nimrod Talmon "Fine-grained liquid democracy"</i>
14.40 -15.20	<i>Gregory Butterworth "Next Generation Liquid Democracy: A.I. Automatic Delegation and Voting Options will Guarantee the Optimisation of Democratic Representative Legitimacy"</i>
15.20-15.35	<i>Break</i>
15.35-16.15	<i>Manon Revel "Liquid Democracy in Practice"</i>
16.15-16.55	<i>Théo Delemazure "Liquid Democracy with Ranked Delegations "</i>
16.55-17.30	Wrap-up of workshop's day 2 and final discussion
From 17.30	End of day 2 and departure of experts and participants

* Nota bene: Marco Deseriis and Julia Wagner could not attend the event on a very short notice (less than a week). Chiara Valsangiacomo (organizers) stepped in to replace Marco and gave a talk about "Six democratic arguments in favor of liquid assemblies."

2.2.3 SCIENTIFIC EXCHANGE OUTPUT – POSTER ABSTRACTS

Liquid Democracy - How Critical are You in Your Network?

Hugo Gilbert (Paris Dauphine University), Rachael Colley (University of Toulouse), Theo Delemazure (Paris Dauphine University)

In this poster, we study, using numerical tests and a theoretical model, the distribution of criticality among voters in a liquid democracy framework. We study a framework in which voters are embedded in a social network and they may vote directly or delegate to one of their neighbors in the network. By criticality, we mean the likelihood that the voter can, by their voting/delegation choice, impact the outcome of the vote. This likelihood relies on a probabilistic model of voters' behavior which, given a voter, depends on their position in the network. Such a probabilistic model defines a probability distribution on delegation graphs, i.e., a graph in which there is one vertex per voter and an arc from voter i to one of her neighbors j if i delegates to j . We design a novel measure of criticality similar to the Banzhaf measure of voting power and show that it can be efficiently approximated by sampling enough delegation graphs. Relying on this result, we focus on the impact of the social network on the criticality distribution among voters. Using various types of graphs (e.g., complete, preferential attachment models, small world), we show how the shape of the network may advantage some voters.

Non-majoritarian Consensus-oriented Liquid Democracy? Combining Probabilistic Voting Rules and Flexible Delegation Systems in the vodle.it App

Jobst Heitzig (Potsdam Institute for Climate Impact Research)

I will present a collective decision-making app that distributes decision power more fairly than existing majoritarian systems and aims for broad consensus. It is based on the idea of conditionally committing to approve options if enough others approve them. This gives a rational incentive to identify good compromise options and agree on them (see working paper). The app will feature a liquid-democracy-style delegation system in its next release. Because the app's ballot requires each option to be assessed independently, delegation will be possible on a per-option basis. In addition, delegation cycles will be resolved in a novel way, by aggregating fallback votes (if any) along the cycle in a certain way. I will discuss the design choices for the delegation component and speculate about its effects on efficiency and the distribution of power.

Constitutive Referendums as Exceptional Democratic Act? Constituent Power, Referendums and the Case of Fusion

Sukmo Kim (University College London)

My presentation is to challenge the normative ground that views specific referendums—a creation of a constitution or a new state—as a direct exercise of the will of the people. My argument is that the referendums require a strong form of mediation by representatives through all stages. The binary image between direct democracy and representative democracy gives an illusion to citizens and becomes a means misguided by the populist.

More specifically, referendums have been considered a direct exercise of the constituent power by the people without a place of representatives. However, I challenge the view of theorizing constitutive referendums as an exceptional democratic act detached from representative democracy. The special moment cannot transfer the plural individuals' will into a singular will of the people. A theoretical model of constitutive referendums requires a specific form of mediation by representatives and revisits the sophisticated referendum process beyond a voting moment. I argue that constitutive referendums can offer a stable form of the democratic process to fabricate specific political preferences and offer narrower guidance to representatives about how to implement those preferences throughout the constitutive change. A fusion case embodies the need for a dynamic interaction between the represented and the representatives.

In the presentation, I try to classify three different types of referendums - policy, constitutional and constitutive - and their normative implications. Then, I explore the reason why referendums should be viewed as a part of the representative system of democracy even in the case of constitutive decisions. In the case of fusion referendums, I set out some difficulties with the referendum process, such as voter qualification and collaborative issues between political parties and civil societies during the stages. Finally, I discuss what safeguard would make a better deliberation and protect minorities during the constitutive referendums, for example - a series of referendums or a super-majority scheme.

I wish that my presentation helps the theoretical gap between the referendum and representative democracy not as a binary but as a spectrum. Even the case of constitutive decisions - secession or reunification - cannot be an exception for a pure form of direct democracy; Instead, the referendums can be instrumentally helpful in giving a specific direction to representatives through a dynamic interaction between the people and the representatives.

Democracy, Decentralization, and Political Evolution: How Technology Enables the Liberal Democracy

Christian Kristensen (Legal Consultant)

My argument has three parts. First of all, I argue that western democracies as we know them today are founded upon a principle of decentralization. Secondly, I argue that complex systems evolve through this very same principle of decentralization, and that this self-organizing mechanism is optimizing the systems adaptability through a process of trial and error. Finally, I argue that liquid democracy is a continuation of "the spirit of democracy", which is to decentralize the power of the state. The fundamental principles of Western democracies. The western liberal democracy is developed from two schools of thought: the religion of Christianity and the ancient Greek philosophy.

Both these systems have the principle of decentralization embedded in them. Before Christ, the law of God came down from the mountain with Moses, and it was the Israelites' duty to follow the commandments written in stone. But Jesus, the divine man, the son of God, the Human being who was God himself, taught his followers that God was within each one of them. That they all—beggars and kings alike—had divine worth, and that everyone was equal in the kingdom of heaven. This means, that not one person—no matter their status—ultimately knows what is right or wrong. This is the principle of decentralization manifested in a religious doctrine. In ancient Greece, the birthplace of democracy, they had a strong tradition for open debate for questioning everything. Everyone today knows the name of Socrates—the most critical and annoying person of all the Greeks. When his friends jokingly asked the Oracle of Delphi, whether Socrates really was the wisest man on Earth, the Oracle did not, as she usually does, give a vague and mysterious as she always does. For once she answered plainly: "yes," and when Socrates heard, his response was that famous phrase that every wise man should learn and

realize: “All I know, is that I know nothing at all”. This is the principle of decentralization manifested as the basis for philosophical exploration. Both Christ and Socrates knew that they alone did not have all the answers. This humble spirit was given a new life during the period known as the enlightenment and was newly formulated in the ideology of classical liberalism—every person is equal before the law – and in the development of the scientific method—we do not know everything, and we can always get smarter. Decentralization as a method of optimized self-organization The principle of decentralization is seen in many institutions of western democracies: In science (the institution of expanding the human bank of knowledge), the peer-review process is crucial, and no scientific theory will ever be able to claim, “absolute objective truth”. It is embedded in the methodology, that every statement should be able to be challenged by anyone, and that theories can only be proved to be false, but never to be true. This method, which emphasizes a scientific process and not an end, has led our knowledge of the material world to grow with exponential speed, leading to all the wonders of human creation we see today. In the capitalist economy (the institution of organizing human labor in a way that benefits all of society in the greatest way possible) the “liberation” of the labor market has meant that every person could choose for themselves what sort of work they wanted to do (within their capabilities) as a contribution to the economy (provided that other people/the economy found their work useful). The productivity of human labor has increased drastically as a result, making us all richer than kings were able to imagine just a few hundred years ago. We know this principle first and foremost as the theory of biological evolution—a theory credited to Charles Darwin. He managed to explain how biological organisms over time optimize their ability to adapt to their environment through a series of genealogical mutations. An organism would reproduce with a slight mutation in its gene-code, and if this mutation increased the organism’s chance of survival and reproduction, it would be passed on through generations and be embedded in the organism. What most fail to mention is that Darwin was reading Adam Smith’s *The Wealth of Nations* as he came upon this idea. What we now call “social Darwinism” was well-known in Scotland (and of course also earlier) before Darwin himself (although many today confuses “survival of the fittest” which speaks to an organism’s adaptability, which can mean basically anything, with “survival of the strongest and most powerful”, which speaks to an organism’s ability to overcome other organisms through the use of brute physical power). The same type of evolution occurs in our scientific exploration and our economic endeavors—we create mutations (new theories or products/ways to use labor—and the more the merrier!) and we test them in their environment (the lab or the free market), and if they are successful they get to grow and become a (more or less) permanent part of the self-organizing mechanism (or complex system, as scholars today would call it). We are even using the same principle in the process of machine learning—in the hope of creating artificial intelligence. Trial and error are how we ourselves learn and develop (why babies put all sorts of stuff into their mouths), and it is how we make complex systems develop over time. A gradual increase will over time lead to an enormous growth – it is what we know as the principle of compound interest. Through a decentralization of the ability to make decisions, we get to try out many more ways (or create many mutations) of doing things, and we increase our productivity massively over time.

Liquid democracy—a decentralization of the power of state Democracy—is already a decentralization of the power of the state. Before democracy, a king ruled as the one who made the rules, the one who enforced them through physical power, and the one who judged those who broke the rules. Democracy is a way of decentralizing this power: First, we separate the power into a legislative, an executive, and a judicial branch. Then we decentralize the legislature and the executive by having elections every four years – increasing the number of “mutations”/changes in which the state is controlled. Liquid democracy is a way to decentralize the state even further. Going back to the principle of the divine individual, every person (with the right to vote) in LD has an equal share in the power to decide how the state ought to use its powers (which are to enforce the monopoly of violence and coercion and spend public means, gathered through taxation, on social services etc.) Here, the individual is given

freedom to set their vote any way they like, and to split their share of power into different political areas (which is an even further decentralization yet!) A decentralization can be beneficial in many public institutions – especially in those which aim to deliver social services such as health care, elder care, and education, in which managerial innovation is necessary for the institutions to adapt to the changing times we live in. From a central hierarchical structure, which the current system still is to some degree, they do not have much room for trying out new ideas and learning from their mistakes. And every time they make a mistake, the central controlling organ (parliament or the executive) must recognize it and legislate according to it and hope that the new piece of legislation will improve the institution. This takes a long time, and LD carries a potential for decentralizing this process—to make sure the public is represented right down to the single school or hospital, and that these can act independently based on trust from those who finance them. Other institutions, such as the state's foreign policy and the defense of its borders, are clearly better suited as a classical top-down hierarchy, so the institutions can act in uniformity. The centralized institution is characterized by uniformity and stability, and the decentralized institution is characterized by dynamical change and innovation. And we need both to have a well-functioning state.

2.2.4 SCIENTIFIC EXCHANGE OUTPUT – PAPER ABSTRACTS

Grassroots vs. Constitutional Liquid Democracy

Ehud Shapiro (Weizmann Institute of Science)

Research in Liquid Democracy typically considers vote delegation to be an integral part of the voting rule. Hence, its protocol should be sanctioned by the constitution (or laws, or bylaws) of the governed entity that employs it. A protocol is termed grassroots if it can have independent deployments that interoperate once interconnected. Here, we propose to consider grassroots delegation as an alternative to constitutional delegation. Grassroots delegation is based on mutual consent between the voter and the delegate; can be deployed independently at different parts of the voting body; can incorporate varying methods of delegated vote aggregation; and, being external to the voting rule, does not require constitutional endorsement.

Grassroots delegation is best thought of as a formalization and a digital embodiment of the informal consultations that take place among family and friends before crucial votes, and of the more formal vote recommendations of political parties and spiritual leaders to their constituencies. Here, we outline a grassroots delegation protocol and compare the intrinsic properties of grassroots liquid democracy and constitutional liquid democracy. Our conclusion is that each has pros and cons compared to the other, but they also have a common core. We propose that research in liquid democracy considers both variants—grassroots and constitutional—and note whether proposals or results apply only to one or to both.

Deliberative Democracy with Dilutive Voting Power Sharing

Dimitrios Karoukis (CEO of Deliberative Technologies)

We present a deliberation model where a group of individuals with heterogeneous preferences iteratively forms expert committees whose members are tasked with the updating of an exogenously given status quo change proposal. Every individual holds some initial voting power that is represented by a finite number of indivisible units with some underlying value. Iterations happen in three stages. In the first stage, everyone decides which units to keep for themselves and where to distribute the rest. With every ownership mutation, a unit's underlying value diminishes by some exogenously given amount. In the second stage, the deliberative committee is formed by the individuals with the most accumulated voting power. These experts can author corrections to the proposal which are proportional to their accumulated power. In the third stage, if an individual outside of the committee disagrees with a correction, she can vote against it with their remaining voting power. A correction is discarded if more than half of the total voting power outside of the committee is against it. If either the committee or the proposal remain unchanged for two consecutive iterations, the process stops. We show that this will happen in finite time.

Is Liquid Democracy Compatible with Representative Democracy? Insights from the Experience of the Pirate Party Germany”

Marco Deseriis (Scuola Normale Superiore)

Although political scientists and political theorists rarely recognize liquid democracy (LD) as a distinct model of democracy, LD has its own history, theoretical underpinnings, and practical applications. The paper fills this gap by conceptualizing LD as an original decision-making procedure and a mode of political representation based on mechanisms of authorization and accountability that are fundamentally different from parliamentary representation. Yet the first practical applications of LD have occurred within representative institutions such as the German Federal Parliament and political organizations such as the Pirate Party Germany. These have experimented with two different LD software, Adhocracy and LiquidFeedback, whose design enables two variants of LD, the former aimed at assessing the quality of opinions and the latter aimed at transforming experts into decision-makers. After examining the impact of the adoption of LiquidFeedback on the internal organization of the German Piratenpartei, the article identifies two challenges that have emerged through the use of LD software: the conflict between the participants, a right to privacy and the transparency of delegated decisions; and the concentration of power in the hands of few delegates. The paper concludes by noting that only the variant of LD oriented toward assessing the quality of opinions is fully compatible with representative democracy.

Nota bene: Marco could not attend the event on a very short notice (less than a week). Chiara Valsangiacomo (organizers) stepped in to replace Marco and gave a talk about “Six democratic arguments in favor of liquid assemblies.” See abstract below.

Six Democratic Arguments in Favor of Liquid Assemblies

Chiara Valsangiacomo (University of Zurich)

The idea of liquid democracy is increasingly being discussed in the academic literature as a democratic innovation that could complement or even replace certain existing democratic practices. Liquid assemblies are a very specific, hypothetical institution through which liquid democracy could be implemented. The present article contributes to this emerging literature by focusing on the following question: Why should we consider injecting liquid assemblies into our democratic systems and which normative problems can they help us solve? Hence, the article analyses the normative value and the possible benefits that liquid assemblies could provide over existing democratic practices. Its aim is to make a theoretical and normative contribution, by presenting the positive arguments that could be employed to legitimize the institution of liquid assemblies and, indirectly, of the broader idea of liquid democracy. Methodologically, the article adopts a problem-based, ecumenical approach to democratic theory. Within this framework, it advances a positive argument in favor of liquid assemblies based on six normative principles of political freedom: political equality, political participation, political representation, deliberation, and quality of outcomes. After presenting a six-point defense of liquid assemblies, it considers two possible objections against this argument. Ultimately, the article suggests that we should consider liquid assemblies as a new and desirable entry into the growing family of democratic innovations.

Online Communities, Misinformation, and Post-truth - A Computational Social Science Perspective

Alessandro Checco (University of Rome La Sapienza)

The online public discourse is under a rapid transformation that is affecting the agency and outreach capacity of social communities and advocacy groups. The loss of a shared system of interpretation of the social reality makes some communities vulnerable to radicalization via data-driven and algorithmic ‘post-truth’ communication techniques that exhaust critical thinking and induce a state of skepticism and isolation. Responses to these challenges have been campaigns to improve digital literacy, critical thinking, fact-checking, and content moderation. However, we argue that these solutions are insufficient as they only consider the recipients of misinformation, overlooking the socio-technical features of the content creation/diffusion process, and disregarding social media platforms’ metrics, architectural features, and algorithmic configurations that contribute to the reduction of dialogue capacity. These issues are particularly damaging for political and advocacy groups that are observing an increasing difficulty in navigating social media communication and engaging the public when developing digital solutions for dialogue and deliberation. This work will discuss a short selection of social science and political philosophy works, read from a computational social science perspective to identify potential issues and blind spots of liquid democracy researchers and practitioners.

Credence Policies and Experts in Democracy

Julia Wagner (ETH Zürich)

Solving complex issues is one of the most important challenges of democratic societies. Many of these issues affect society in several dimensions and are tried to be solved by implementing policies. In the case of credence policies, short- and mid-term consequences of their implementation are often difficult or even impossible to predict because they are highly uncertain. Moreover, they remain unknown for quite some time. Examples are managing a health crisis, slowing down climate change, or imposing banking regulations. In any of these cases, experts can provide information about the consequences of policies, new or past. From a democratic perspective, the role of experts is important as they can influence policy decisions, citizens’ attitudes towards them and hence indirectly election outcomes through their opinions.

There are different ways for a society to include expertise in decision-making processes. Given that a part of the electorate is willing to listen to experts’ opinions, there are two well-known procedures for the inclusion of experts in the decision-making process. First, one can use Vote Delegation: voters can delegate their votes to representatives. Second, one can rely on Opinion Updating: experts first reach a consensus and then make it publicly available to voters before voting day. In this paper, we analyze and compare the two voting procedures, Vote Delegation and Opinion Updating to the simple majority vote without experts and to an optimal policy to assess the ideal design of democracy for complex policy issues. For this purpose, we use a theoretical model with heterogeneous voters—who form the electorate—deciding on the extent of a credence policy. This is a policy whose benefits cannot be observed, neither in the short run nor in the long run. The voters differ in two dimensions, risk aversion and beliefs about the efficiency of the credence policy (alternatively, the costs associated with the credence policy). While the beliefs of some voters may change in response to the experts’ actions, the voters’ risk aversion is fixed and invariable. Yet, one part of the electorate is opinionated, meaning

that it cannot be influenced by experts' opinions. For simplicity, experts are modeled by one representative expert, whose opinion represents the experts' consensus about the efficiency of the credence policy. We assume in this paper that experts with different opinions discuss policy recommendations and publish one consensus. The main part of the paper deals with the analysis of the policy outcomes of the four different voting procedures. We show that regular elections always produce executive action, and the chosen policy is higher for larger shares of opinionated voters.

We further find that experts can have a beneficial, neutral, or harmful impact on democratic decision-making. Extreme and detrimental outcomes can arise when decisions are delegated to experts who act on the most accurate assessments of the consequences of policies but do not consider differences in risk aversion in the electorate. Publishing a consensus assessment of experts before citizens vote can be neutral or can improve policymaking depending on how beliefs about the effectiveness of the policy are distributed in the electorate. Furthermore, by considering a convex combination of the utilitarian and maximin welfare, we can prove that there always exists an interval of welfare-optimizing policies. In conclusion, experts' influence on policies and welfare depends on how they are included in the decision-making process.

Nota bene: Julia could not attend the event on a very short notice (less than a week). Her talk was therefore cancelled.

Electoral Market Structure under Liquid Democracy

Dylan Johnson (University of Hamburg)

While its intellectual origins can be traced back to Lewis Carroll's 1884 book, *The Principles of Parliamentary Representation*, (Carroll 1884) Liquid Democracy (LD) has more recently received significant, and growing, attention from academic researchers and political parties (Paulin 2020). This flood of interest in LD can primarily be attributed to two factors—its theoretically demonstrated advantages vis-à-vis representative and direct democracy with respect to certain epistemic (Ford, 2020; Green-Armytage, 2015; Kahng et al, 2021) and normative criteria (Blum & Zuber, 2016; Landemore, 2020; Valsangiacomo, 2021), and the advent of new technologies that have finally rendered its mechanics technically feasible. While the existing literature provides ample reason for researchers and policymakers to seriously consider full or partial implementations of LD as viable mechanisms for improving democracy, progress on this front is limited by the relative paucity of analyses considering how institutional design choices may affect the incentive structure facing voters and proxies. In seeking to help close this gap, this paper makes two primary contributions to LD research. First, it argues that ideal conceptions of LD closely parallel those of a perfectly competitive market—a claim that, if true, opens this subject to several fruitful lines of economic analysis. Accordingly, it then considers the importance for LD of accounting for an efficient pricing mechanism that does not undermine the political equality of voters—a problem for which it puts forward a novel solution. Regarding its first contribution, it adopts a Public Choice-inspired understanding of electoral systems as a type of market, wherein voters (consumers) seek to maximize their preference satisfaction through exchange with similarly motivated proxies (suppliers) under conditions of scarcity. Voters wish to maximize the degree to which votes cast on agenda items are in line with their own preferences—a product they can either “purchase” from a proxy or “produce” themselves (in a LD). When “purchasing” this “product” from proxies, they “pay” their counterpart by transferring to them (all or part of) their finite voting allotment—where proxies, in return, seek to maximize their “profit” measured as the difference between total votes acquired and the expenses incurred to receive these votes. Through this lens, when one

considers electoral markets as they appear in representative democracies (RD), substituting “political candidates” or “parties” for “proxies”, it is easy to recognize the supply-side problems that emerge in RD. Namely, due to both constitutional features and practical concerns that effectively constrain the number of parties that may exist within a RD system, electoral markets therein are all but destined to become oligopolistic – meaning, a market structure in which “consumers” (i.e., voters) can only choose from a greatly constrained set of products offered by a limited number of “suppliers” (i.e., “parties”). Thus, these constraints on voter choice create a market failure where a large portion of voter preferences remain unsatisfied. However, because supply-side competition is limited and RD awards substantial benefits to electoral winners, parties and candidates are motivated to bring “products” to the market by the allure of capturing oligopoly rents through winning public office. LD, on the other hand, need not result in an oligopolistic electoral market. Its central institutional features place, in principle, no a priori barriers to entry and exit on the market (insofar as every voter can freely choose to become, or stop being, a proxy). Moreover, combining “instant recall” with the assumption of perfect information allows for perfect elasticity of demand, as voters may easily adjust their delegation decisions (i.e., purchasing behavior) in response to changes in supply. This should necessarily be the case given that, due to instant recall, there should always exist for every voter at least one proxy who could provide a perfect substitute for their desired “product”, because if no other proxies are available on the market, individuals could select themselves as their own proxy. Therefore, voters could freely choose between suppliers offering homogenous products (viewed as the service of voting on agenda items according to defined preference dimensions), and each supplier should have no ability to unilaterally affect “prices”, as voters could redelegate either to themselves or another proxy providing the same service. Accordingly, insofar as these assumptions hold regarding the core features of LD, it appears formally capable of permitting the emergence of a perfectly competitive electoral market.

The analytical tool of perfect competition is useful for researchers not as a description of how markets function, but as a tool for identifying which factors serve as important constraints on their ability to meet this ideal in practice. Thus, it seems that LD research would similarly benefit from this type of analysis. This paper’s second contribution, then, works in this vein. It contends that a logical implication of considering electoral systems under LD to be potentially perfectly competitive electoral markets is that they must necessarily contain an effective pricing mechanism. This is because prices are indispensable for markets, as they communicate dispersed information and incentivize the production and consumption behavior that enables markets to function efficiently. However, there is little analysis in the literature of how such a mechanism could exist within a LD electoral market, much less how it could simultaneously preserve the political equality of voters. This problem is frequently hinted at within the literature, but is often either downplayed (i.e., through the argument that sufficient proxy supply could be achieved via the existence of intrinsically motivated, “gyroscopic” proxies”) (see Valsangiacomo, 2021) or left as a topic for future research.

One exception to this is Green-Armytage’s (2015) short treatment of a remuneration system for proxies, wherein a yearly vote would be held to determine which proxies should receive a given proportion of a collective pool of funds designated for this purpose. This proposal is interesting, in that it appears to be capable of preserving the political equality of voters while still permitting some form of pricing system to operate. However, the extent to which it would perform these functions remains unclear. Namely, its pricing mechanism appears inefficient, as it seems likely to render supply more price inelastic because the financial incentives tied to LD’s “instant recall” function would be lagged. Also, while permitting voters to allocate heterogeneous proportions of their assigned share of the collective fund to different proxies may communicate more information about voter preference intensity, it may undermine political equality. This is because voters would not be uniform in their ability to financially incentivize proxies across all relevant policy dimensions through the use of public funds. Thus, the second part of this paper puts forward a modified version of Green-Armytage’s

remuneration scheme that attempts to resolve this issue. Essentially, it describes a system in which voters are each endowed by the state with an equal vote allocation, which is in turn decomposable into smaller voting units equally distributed within a finite, standardized set of policy dimensions. Each voting unit is then redeemable by proxies for an equal sum of currency, paid out from a collective fund in regular intervals. Thus conceived, this proposal may constitute a pricing mechanism that communicates information about voter preferences and incentivizes proxies to supply services that satisfy these preferences, while doing so in a manner that preserves the political equality of voters.

Fine-grained Liquid Democracy

Nimrod Talmon (Ben-Gurion University)

I will speak about Fine-Grained Liquid Democracy (FGLD) where voters can partially delegate their votes to different delegates. I will survey my work (with co-authors) on FGLD for ordinal elections, knapsack voting, and cumulative ballots. I will argue that, even though allowing such partial delegations can be computationally extensive, algorithmic techniques can be applied to make it possible to harvest the benefits of such expressive forms of delegations while still being efficient and easy-to-use.

Next Generation Liquid Democracy: A.I. Automatic Delegation and Voting Options Will Guarantee the Optimisation of Democratic Representative Legitimacy

Gregory Butterworth (Cardiff University)

Dissatisfaction with representative democracy has led to a search for superior systems of government. There is growing interest in utilizing the internet to evolve democracy towards mass participation via propositions such as liquid democracy, which is a hybrid direct-representative decision-making process that shifts the decision-making power to the electorate. The central concept of liquid democracy is that, on a per issue basis, each voter has the option to either vote directly or to delegate their vote to a more informed voter, hopefully resulting in better decisions. Note that votes are transitive, i.e., an agent that receives delegated votes may in turn delegate those votes to another agent. However, the acceptability and uptake of the liquid democracy is hampered by various theoretical apprehensions. Apprehensions include, but are not limited to the power of users that have amassed many delegated votes; the failure of assumed delegation to the more informed to collectively arrive at better decisions than direct or representative democracy; voting security; democratic representative legitimacy; etc. Therefore, the widespread adoption of liquid democracy is hampered until it can allay such concerns and can be shown to be the most obviously preferable alternative to the democratic system that it intends to replace. From the above apprehensions we focus on the topic of democratic representative legitimacy. We propose that a next-generation liquid democracy system with A.I. based automatic delegation and voting options creates a better user experience, and optimal democratic representation. More specifically, for those agents that do not have the time to vote or find a suitable delegate, we propose a next-generation liquid democracy system that implements machine learning functionality via agent supplied political profiles to provide agent similarity based automatic voting and delegation options. We then argue that the above functionality will, at least from a practical sense, lead to optimal democratic representation. The intended audience is wide ranging and aims to bridge a gap between political and computer scientists. Therefore, in an attempt to keep this article suitably accessible and self-contained, it includes for the benefit of computer scientists, a rudimentary introduction to the concept of democratic representative legitimacy, and for the benefit of political scientists, a rudimentary introduction to machine learning.

Liquid Democracy in Practice

Manon Revel (Massachusetts Institute of Technology)

Proposals to change the size of the Congress, update ballot types, channel decisions through citizens' assemblies or participatory budgeting are being actively researched and tested. Liquid democracy is making its way to the democratic innovation table as a model that may stand for non-active stakeholders and enhance collective intelligence using the wealth of inter-personnel information embedded in social networks. Yet, much uncertainty remains about its propensity to concentrate voting power in the hands of a few, its likelihood to ease vote buying, or to create problematic (e.g., cyclical) delegations. In this talk, I will review what we currently understand theoretically and empirically about liquid democracy and interpret how that informs further real-world testing and deployment of liquid democratic procedures.

Liquid Democracy with Ranked Delegations

Théo Delemazure (Paris Dauphine University)

Liquid democracy is a novel paradigm for collective decision-making that gives agents the choice between casting a direct vote or delegating their vote to another agent. We consider a generalization of the standard liquid democracy setting by allowing agents to specify multiple potential delegates, together with a preference ranking among them. This generalization increases the number of possible delegation paths and enables higher participation rates because fewer votes are lost due to delegation cycles or abstaining agents. To implement this generalization of liquid democracy, we need to find a principled way of choosing between multiple delegation paths. In this paper, we provide a thorough axiomatic analysis of the space of delegation rules, i.e., functions assigning a feasible delegation path to each delegating agent. We prove axiomatic characterizations as well as an impossibility result for delegation rules. We also analyze requirements on delegation rules that have been suggested by practitioners and introduce novel rules with attractive properties. By performing an extensive experimental analysis on synthetic as well as real-world data, we compare delegation rules with respect to several quantitative criteria relating to the chosen paths and the resulting distribution of voting power. Our experiments reveal that delegation rules can be aligned on a spectrum reflecting an inherent trade-off between competing objectives.

2.2.5 SCIENTIFIC EXCHANGE OUTPUT – PICTURES



Day 1 – Poster Session



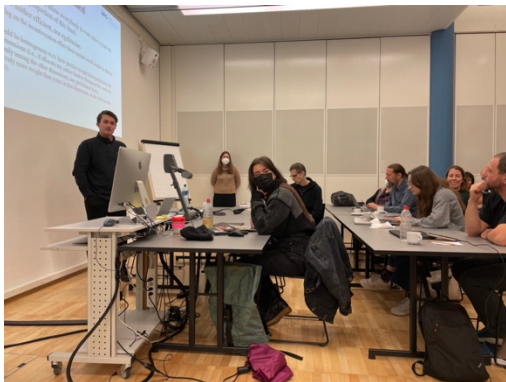
Day 1 – Panel I (Alois Paulin)



Day 1 – Panel I (audience)



Day 1 – Public roundtable



Day 2 – Panel II (Dylan Johnson)



Day 2 – Panel III (Manon Revel)

2.3 OUTPUT DATA FROM THE WORKSHOP

All the output data from the liquid democracy workshop were all listed on mySNF (i.e., academic event, collaboration, and third-party funds).

3. PARTNERSHIP ASPECTS

3.1 ADDED VALUE FOR SWISS SIDE

This workshop on liquid democracy comes at troubling times for democracies. We hope that fostering connections between scholars and practitioners will contribute to a better understanding of the problems at hand that liberal democracies in the West are facing and to proposing impactful solutions to democratic decision making. With this workshop, our intention was to place the co-organizers from Switzerland together with their institutions at the forefront of the international academic research of this new and timely topic that is liquid democracy. With the unique combination of a rich and strong political culture together with the interdisciplinary research fields at the University of Zurich, we were able to promote our respective research fields of political science, theoretical mathematics, normative democratic theory, ethics, and computer science. The workshop offered us the opportunity for networking and allowed establishing new contacts and set the groundwork for future collaborations.

3.2 FURTHER COLLABORATION

Over the course of the workshop, many ideas around how to face challenges and how to advance the field of liquid democracy came up. The ideas floating around are numerous. For the context of this report, however, we would like to focus on the collaboration that are a.) at a more mature stage (i.e., goal and/or outline were identified) and b.) involve workshop co-organizers from the University of Zurich. Given these two criteria, we want to highlight the following collaborations:

- Christina Zuber and Chiara Valsangiacomo: Organizing a scientific research exchange for the postdoc career of Chiara Valsangiacomo.
- Karsten Donnay and Lucien Heitz: Using/organizing a household panel to survey and measure how technology impacts political participation (i.e., extend the existing research platform *Informfully* of Lucien Heitz to regularly ask people how, e.g., online news, impacts their voting behavior).
- Ehud Shapiro, Cristina Sarasua, and Lucien Heitz: Plans for establishing of a NPO in Switzerland to develop and promote a decentralized online platform for deliberation and decision-making. Ehud Shapiro has already founded many successfully businesses and a political party. For the newest endeavor on grassroots liquid democracy, he was looking for international partner/partner institutions in Switzerland.
- Bryan Ford, Davide Grossi, and Lucien Heitz: Collaborations for development of “normative algorithmic system” that focus generating value for society. E.g., curation algorithms for online news recommender systems, that not only focusing on economic incentives, but also address the information needs of citizens in democratic societies when creating recommendations for users.

4. APPENDIX

4.1 FUNDED PARTICIPANTS

The funds from the scientific exchange grant were intended for travel and subsistence allowances for a total of three international workshop collaborators: 1.) Manon Revel, doctoral researcher at the Institute for Data Systems, and Society, at the Massachusetts Institute of Technology (USA), 2.) Davide Grossi, professor at the Faculty of Science and Engineering, at the University of Groningen (The Netherlands), and 3.) a non-named collaborator spot for either the expert speaker Andreas Nitsche (Association for Interactive Democracy, Berlin, Germany) or Markus Strohmaier (professor at the University of Mannheim, Mannheim, Germany). Unfortunately, both options for the non-named collaborator position had to cancel their participation in the workshop. Thankfully, we were able to bring on board Karsten Donnay (professor at the University of Zurich, Zurich, Switzerland) to fill the expert position. Manon Revel and Davide Grossi were able to participate in the workshop as planned. Furthermore, with the generous grant of scientific research program, we were able to secure additional funding. We received a Short Grant from the Graduate Campus of the University of Zurich.

In front of the background of these circumstances, we would like to briefly comment on the subsequent changes we had to make with respect to the financial report. We stayed well below our initial budget and did not use all the allocated funds. The expenses for Manon Revel for travel and subsistence allowances remained unchanged. The costs for hosting Davide Grossi, however, were covered by the Graduate Campus grant instead. Furthermore, with Karsten Donnay filling the non-named expert position, no additional funds had to be used to host him (Karsten Donnay is from Zurich and had no additional travel or accommodation expenses). In summary, the funds had to cover only one of the initial three participant slots.

4.2 ACKNOWLEDGEMENTS

This workshop could not have been organized without the financial support provided by the following institutions:

- the Graduate Campus via a Short Grant (CHF 3,790)
- the Swiss National science foundation via a Scientific Exchange Grant (CHF 3,600)
- the Doctoral Program Democracy Studies, which writing advertisement material (congress set with A4 folder, notebook, and pencil) for all participants and offered the lunch to all participants on Day 2.
- The Digital Society Initiative of the University of Zurich via the Excellence Program Scholarship of Lucien Heitz.



4.3 PROJECT LEADERS

- Lucien Heitz, Doctoral researcher, University of Zurich, Department of Informatics
- Manon Revel, Doctoral researcher, Massachusetts Institute of Technology
- Cristina Sarasua, Postdoc researcher, University of Zurich, Department of Informatics
- Chiara Valsangiacomo, Doctoral researcher, University of Zurich, Department of Political Science